

SIT215 Computational Intelligence

Assignment 3:

Integrated Intelligent System Project



Grading and weighting: (% total mark for unit): 40%

Submission Deadline: 8:00 pm AEST, 31 May 2026 (Week 12 Sunday)

Learning Objectives

This assignment focuses on implementing and evaluating a fuzzy logic foundation, then integrating it into a hybrid computational intelligence system. It is designed to enable you to demonstrate your learning achievements in ULO1, ULO2, and ULO3 of this unit, including:

- Developing a functional and technically sound intelligent system for a defined problem
- Implementing, testing, and integrating computational intelligence techniques in code
- Evaluating system performance with attention to robustness, ethical implications, and cybersecurity considerations
- Communicating design decisions, implementation logic, testing methodology, and evaluation outcomes clearly and professionally in written and oral formats

Assignment Overview

In this assignment, you will build an integrated intelligent system for a goods delivery routing problem. Throughout the trimester, you have discussed and practised important computational intelligence terms and ideas such as agents and environment, state-space representation, A* planner, admissible heuristics, fuzzy membership functions, rule bases, fuzzification, defuzzification, hybrid reasoning, replanning, robustness, and responsible system design. This assessment asks you to bring those ideas together in one coherent project.

Problem:

A delivery company needs to transport goods between two locations (Node A to Node B) on a road network. Each road segment has a known **distance** (in km), a **bumpiness** level described in linguistic terms, and the goods being delivered have a **fragility** rating, also described linguistically. Based on the bumpiness of each road segment and the fragility of the goods, a fuzzy logic system determines the **maximum safe speed** for each segment. **The goal of a hybrid computational intelligent system for this problem is to find the fastest route** two locations (Node A to Node B) using A* search with integrated fuzzy logic. Then, the hybrid system must **also respond to a changed environment through replanning**.

How can a fuzzy inference system be implemented, evaluated, and integrated into a hybrid computational intelligence system that produces explainable routing decisions in a changing delivery environment?

Your solution must demonstrate:

- environment modelling and abstraction
- A* search design and execution
- fuzzy logic system design and justification
- hybrid integration of search and fuzzy reasoning
- replanning under changed environmental conditions
- evaluation of the performance and examine responsible and rational behaviour

These tasks integrate concepts across agents, fuzzy systems, search, evaluation, and responsible system design. You are expected to deliver a working solution, justify your design choices, and explain why the integrated behaviour is intelligent and technically sound.

Submission

This is an individual assessment task. Please read this assignment tasksheet carefully. You **must** submit the following files to the relevant Assignment Dropbox folder on the Unit Site no later than the given deadline:

- 1) **The main report:** *[YourID]_assign3integratedsystem_report.pdf*. Your report must clearly present your solution, design decisions, testing approach, and evaluation. Every submission must follow the standardised report structure, outlined in section [Additional Guide for You to Complete this Assignment Effectively](#) of this tasksheet.
- 2) **Source code: Jupyter Notebook** (*[YourID].ipynb*) **or Python files** (*[YourID].py*). All codes were developed or used as your complete solution in this assessment. Your code must be in appropriate documentation with sufficient comments, explaining the key logic of your implementation.
- 3) **README file:** *[YourID]_README.txt* **or** *[YourID]_README.md* with clear execution instructions, specifying any required libraries or dependencies.
 - Note: If your code solution includes the usage of any open-source libraries, you need to provide those libraries or make them accessible to the assessor (e.g., by providing a link to a download site and instructions on how to install and use the library in your solution).
- 4) A 3-minute narrated demonstration video - **.mp4 format:** Upload your own local video file as *[YourID]_demo video.mp4* directly to the Assignment Dropbox **or** use [Deakin Panopto](#) to create and upload your video)
 - Note 1: It is the student's responsibility to ensure the video is accessible.
 - Note 2: Submission via a link to any web video-hosting platform will not be accepted. Only video files uploaded directly to the unit site Assignment Dropbox are recognised as valid submissions.

Late Penalties

In accordance with Faculty assessment policies, late submissions to the submission folder will incur a penalty based on Deakin's Penalties for late submission and Due dates policies, <https://www.deakin.edu.au/students/study-support/assessments-and-examinations/assessments>

Your Tasks

An Assignment 3 submission must present a complete and coherent integrated intelligent system solution. All Tasks 1–4 are compulsory and will be evaluated together as one hybrid computational intelligence project.

Task 1: Baseline A* planner with constant speed

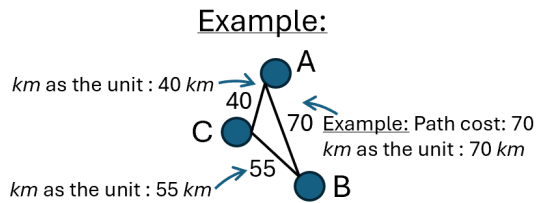
You are required to establish the baseline routing problem and solve it using A* search with a constant speed assumption. This baseline will later be used to support comparison with the integrated hybrid system.

1.1) Model an environment using an appropriate representation

- You may reuse and refine your state-space graph from Assignment 1 where appropriate. Otherwise, create your environment (from real-world map with connected roads) to address the given problem
- Your environment model must:
 - include 35-38 meaningful path segments among 20 – 23 nodes, in the state-space
 - clearly define nodes and paths, and associated path costs

1.2) Define a relevant cost function for the baseline case.

- Firstly, *as an intermediate step*, define distance (in km) as the unit for the path segment cost, see below illustration



- Then, use a constant travel speed of 100 km/h for each of the path segment, define a relevant cost function for this baseline case.
 - Hint: The goal of this stated problem is to find the fastest route from a defined start node to a defined goal node using A* search.

1.3) Implement A* search for the baseline constant-speed case.

- Define an appropriate heuristic for this problem setting.
 - Hint: For example, using Euclidean distance and the constant travel speed of 100 km/h to formulate the heuristic
- Execute A* and show the full search trace, including actual path cost $g(n)$, heuristic cost $h(n)$, total cost $f(n)$, on nodes to explain how the final path is produced.

1.4) In your report sections: *Baseline A* planner with constant speed*, include the following (but not limited to)

- Describe your routing problem and show the state-space graph
- Explain the baseline cost function and the admissibility of the heuristic
- Explain the resulting path using the above-stated A* full search trace information
- Present the resulting path, total distance, and total travel time for the constant-speed case

Task 2: Fuzzy inference system design

You are required to implement and evaluate a fuzzy logic foundation that determines safe travel speeds for path segments. This task establishes the reasoning component that will be integrated with the replanning solution in Task 3.

2.1) Design the fuzzy Inference system (You may reuse and refine relevant elements from Assignment 2 where appropriate.)

- Define (or reuse the deliverable from Assignment 2), the fuzzy input variable fragility with an appropriate universe of discourse and three membership functions.
- Define the fuzzy input variable bumpiness with an appropriate universe of discourse and three membership functions.
- Define the fuzzy output variable maximum safe speed with an appropriate range and three membership functions between 40 km/h and 100 km/h.
- Design a complete and logically consistent fuzzy rule base that covers all input combinations.

2.2) Implement the fuzzy inference system

- Complete the fuzzy inference process, including fuzzification, rule activation, aggregation, and defuzzification to output ranging between 40 km/h and 100 km/h.

2.3) In your report section: *Fuzzy inference system design*, explain the following (but not limited to)

- With the aid of membership functions plot(s) and Rule Matrix, explain the logic of your rule base and justify how it is appropriate for the problem.
- With the aid of the defuzzification output plot, show how the suggested crisp safe speed is produced for one representative path segment.
 - Note: Your implementation must be sufficiently documented so that its behaviour can be understood without step-by-step debugging.

Task 3: Integrated intelligent system, replanning, and responsible evaluation

You are required to integrate the fuzzy logic foundation into a hybrid computational intelligence system and evaluate its performance in a changing environment. This task is the core of the assignment and must demonstrate successful integration, testing, and explanation of the resulting system.

3.1) Design and implement the integrated system

- Use the defuzzified speeds produced in Task 2 to refine the A* search model from Task 1.
- Define and justify an appropriate heuristic informed by the fuzzy inference system for the integrated case, provided it remains an estimate that does not overestimate the true remaining cost.
- Execute A* with the fuzzy-informed heuristic and evaluate the resulting path across several fragility levels respectively, comparing its performance with the outcome of the Task 1 baseline case.

3.2) Replan under a mid-journey environmental change

- Assume that, after 20% of the originally planned route, measured by the number of nodes visited and rounded up to the nearest whole node, the school-zone condition begins. From there, 60% of the path segments in state-space are constrained by a 40 km/h speed cap.
 - The 60% of path segments must be selected using a fixed and clearly documented method (e.g., a predefined subset or a reproducible random seed) and reused consistently across all comparative experiments.
- Design, implement, and execute a reactive layer to make replanning decisions between two cases. For school-zone-constrained path segments, the travel speed become 40 km/h. For unconstrained segments, the effective travel speed remains the fuzzified safe speed.
- Compare the planned path before replanning with the actual path after replanning, including differences in route, total distance, total travel time, and the effect of replanning under different fragility levels respectively.
 - Hint: Replan from the current node rather than from the original start node.

3.3) In your report *section: Integrated system results and evaluation*, include the following (but not limited to)

- Provide a block-based flowchart that illustrates the reactive layer of your agent-environment loop. The flowchart must clearly show how fuzzy logic is incorporated and represent key capabilities and components of the agentic system. The flowchart annotations correspond directly to your class names and method names in code.
- Justify why and why not, the fuzzy-informed heuristic in the integrated system may remain admissible, and compare it with the heuristic used in Task 1 to evaluate which is more effective under a more constrained problem setting.
- Compare the original planned path before replanning with the actual travelled path after replanning, including changes in route, total distance, total travel time, and the effect of replanning across different fragility levels.

Task 4: A 3-Minute Video Presentation

Create a narrated overview of your integrated intelligent system while displaying the relevant contents of your implementation outcome / deliverables on the screen, in a 3-4 minute video. Your video should:

- highlight the key goal(s) of the problem and the intended outcome(s),
- explain how multiple reasoning logic had been applied in your integrated system, and
- identify the constrained sub-problems you have implemented and tested, describe the performance of the system focusing on how its applied intelligence supports rational decision-making.

The video is intended to be concise. Videos that significantly exceed the expected duration may affect the mark for presentation and communication if the explanation becomes unfocused, repetitive, or fails to communicate the required points efficiently.

Submit your video file to the Assignment 3 Dropbox folder, along with all other required files. Ensure that your video is clear, structured, and accessible for assessment.

Marking

Your submission will be evaluated based on criteria that reflect the depth of your problem-solving skills, integration quality, technical proficiency, theoretical knowledge, and communication effectiveness against the

quality of your work. A numeric mark (out of a total mark of 40) will be determined by how well your work aligns with the grading criteria and its standard as outlined in the rubric available on the Unit Site under Resources>Assessment.

Additional Guide for You to Complete this Assignment Effectively

Preparatory Learning Activities

To complete this assessment task, you will need to first develop an understanding of a range of topics covered in this unit across

- agents and environment
- A* search and heuristic admissibility
- fuzzy membership functions, rules, and inference
- hybrid computational intelligence integration, adaption and replanning consideration.

Understanding these areas and how they inter-relate will make it far easier to understand the learning materials, and explain your investigations and outcomes in this assignment. Our advice is that you use this assignment as a basis for the study of these underlying areas to assist in integrating the knowledge covered in this unit into a meaningful ‘whole’, which supports completing this assessment task.

Report Structure

You may reorder the following sections where appropriate to reflect the logical progression of your work.

Sections	Guide for you															
Cover Page	Must provide checklist below to ensure clarity in your submission. Tick (✓) the boxes next to the tasks you completed to help you verify your work before submission.															
	<table><tr><th>Tasks</th><th>Completed (✓)</th><th>Notes (Optional)</th></tr><tr><td>Task 1: Baseline A* planner with constant speed</td><td>☐</td><td></td></tr><tr><td>Task 2: Fuzzy inference system design</td><td>☐</td><td></td></tr><tr><td>Task 3: Integrated intelligent system, replanning, and responsible evaluation</td><td>☐</td><td></td></tr><tr><td>Task 4: Recorded video presentation submitted</td><td>☐</td><td></td></tr></table>	Tasks	Completed (✓)	Notes (Optional)	Task 1: Baseline A* planner with constant speed	☐		Task 2: Fuzzy inference system design	☐		Task 3: Integrated intelligent system, replanning, and responsible evaluation	☐		Task 4: Recorded video presentation submitted	☐	
	Tasks	Completed (✓)	Notes (Optional)													
	Task 1: Baseline A* planner with constant speed	☐														
	Task 2: Fuzzy inference system design	☐														
	Task 3: Integrated intelligent system, replanning, and responsible evaluation	☐														
Task 4: Recorded video presentation submitted	☐															
Baseline A* planner with constant speed	Establish the routing problem, the environment representation, the baseline cost model, the justification of the heuristic, and the resulting path. See Task 1.4 for further details.															
Fuzzy inference system design	Explain the fuzzy variables, membership functions, rule base, representative inference outcomes, and the consistency of the fuzzy logic framework. See Task 2.3 for further details.															
Integrated intelligent system, replanning, and responsible evaluation	Explain the system integration, replanning under changed environmental conditions, performance evaluation, responsible system considerations, and the flowchart annotations aligned with your code. See Task 3.3 for further details.															
Conclusion	Summarise the main findings, relate them to the evaluation results, and identify the scenario in which hybrid reasoning is most needed.															
Acknowledgement	Record any external assistance, if applicable.															
References	List all citations and resources used, using an appropriate academic referencing style.															

Expected Report Length

Your report should be concise yet comprehensive, focusing on the key components of your solution while maintaining depth and coherence. Where relevant, use appropriate visuals (e.g., maps, graphs, tables, figures, or flowcharts) to support your analysis. All sources must be cited and referenced using a consistent academic referencing style. If completed correctly, *as a guide*, the expected length is approximately 7 to 10 single column A4 pages, using font size of 11 with single line spacing, covering all essential sections as the outlined.

Report Structure:

- Cover Page: include only the required checklist table, along with your name and student ID
- Baseline A* planner with constant speed: 2-3 pages
- Fuzzy inference system design: 1-2 pages
- Integrated intelligent system, replanning, and responsible evaluation: 4-5 pages with code-aligned annotated flow chart
- Conclusion: approximately 50 words
- Acknowledgements (if any) and References: no strict page limit

Getting Help and Support

Students are encouraged to collaborate by discussing tasks and assisting each other in understanding concepts, models, and algorithms. Engaging in class sessions (Lecture, Prac) and participating in the unit site discussion forum or Teams support channel can strengthen your understanding and contribute to peer learning. However, this is an individual assessment, and all development work and report writing must be your own. If using external references, ensure they are properly cited. Collusion and plagiarism will result in severe academic penalties. If you need clarification on completing this task, seek guidance from the teaching team.

Guidance Support

The teaching team is available to assist you with this task. If you need help understanding the material or requirements, seek support during class time or post your questions in the Discussion Forum on the Unit Site, where responses can also benefit your peers.

Report Writing Help

While the teaching team can offer guidance on report writing, ensure you review the instructions in this task sheet carefully. Additional support is available through the Writing Mentors program, which provides assistance with academic writing, including report preparation. For more information, visit <http://www.deakin.edu.au/students/studying/study-support/writing-mentors>.

Feedback

Students are encouraged to actively consolidate their design inspiration from peer discussion in class, and seek formative feedback from teaching staff on their work completed before the submission deadline to ensure they are on track with this task. Formative feedback may be obtained during weekly scheduled practical classes upon request. Talk to us, and we'll support you! Besides, in 15 business days after the assignment submission due date, marking feedback will be provided via the unit site.

End of Assignment 3 Tasksheet.